

Performance Test

Test Run ID: be7f2a7e-4b30-46b4-a392-7e6dbd90f3ec

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	05 June 2026	Max VUs Tested	250
Executor	ramping-arrival-rate	Tool	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a load test, utilizing a ramping arrival rate scenario to simulate a gradual increase in traffic. The load profile consists of a single stage with a maximum of 50 Virtual Users (VUs) and a target rate of 250 requests per second (RPS) sustained for 30 seconds.

Stable Threshold	Degradation Onset	Critical Point	Peak Throughput
37-81 TPS	N/A	N/A	81 TPS

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	258.94ms	1056.36ms	232.95ms	217.18ms	373.01ms	409.84ms	-
http_req_waiting	257.95ms	588.30ms	232.61ms	216.62ms	371.71ms	408.18ms	-
http_req_connecting	9.91ms	249.43ms	0.00ms	0.00ms	0.00ms	0.00ms	-
iteration_duration	1280.16ms	2058.11ms	1234.93ms	1218.93ms	1405.61ms	1476.47ms	-

Counters

metric	count	rate
data_received	3.97 MB	130.26 kB/s
data_sent	145.65 kB	4.66 kB/s
http_reqs	1,167	37.37/s
iterations	1,167	37.37/s

Rates

metric	rate
checks_succeeded	100.00% (1167/1167)
checks_failed	0.00% (0/1167)
http_req_failed	0.00% (0/1167)

Gauges

metric	value
vus	6 (min: 6, max: 50)
vus_max	50 (min: 50, max: 50)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	1167	0	100.00%	Pass

3. Advanced Analysis

3.1 Error Analysis

- **Network/Connection errors:** None detected.
- **Application errors:** None detected.
- **Infrastructure errors:** None detected.

- **Insufficient VUs:** The test reached 50 active VUs and was unable to initialize more, resulting in a warning message. This is not an error, but rather a limitation of the test setup.

3.2 Observation

The test results indicate that the system was able to handle 50 active VUs without any significant issues. However, the warning message about insufficient VUs suggests that the test was not able to scale to the desired level. This could be due to various reasons such as:

- **Client timeouts:** The `http\_req\_duration` metric shows an average duration of 258.94ms, which is relatively high. This could indicate that the clients are taking a long time to receive responses from the server, potentially due to network latency or server-side processing delays.
- **503 Service Unavailable:** Although not explicitly mentioned in the logs, the high `http\_req\_duration` and `http\_req\_waiting` metrics suggest that the server may be experiencing high load and returning 503 responses to clients.
- **Session not found:** The `dropped\_iterations` metric indicates that 3333 iterations were dropped, which could be due to session not found errors or other issues with session management.
- **Silent 500 session initialization failures:** The `dropped\_iterations` metric also suggests that some iterations may have failed silently due to session initialization issues.

3.3 Final Recommendations

1. **Advanced Connection Management:** Implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.
2. **Session Persistence sticky affinity/Redis cache:** Implement session persistence using sticky affinity or Redis cache to reduce the number of session initialization failures.
3. **Gateway resilience Istio outlier circuit breaker:** Implement Istio outlier detection and circuit breaker to detect and prevent cascading failures in the gateway.
4. **Observability tracing and standardized log levels:** Implement observability tracing and standardized log levels to better understand the system's behavior and identify issues.
5. **Database read/write splitting and PgBouncer connection pooling:** Implement database read/write splitting and PgBouncer connection pooling to reduce the load on the database and improve performance.

3.4 Conclusion

The system was able to handle 50 active VUs without any significant issues, but the warning message about insufficient VUs suggests that the test was not able to scale to the desired level. The high `http\_req\_duration` and `http\_req\_waiting` metrics suggest that the server may be experiencing high load and returning 503 responses to clients. The `dropped\_iterations` metric indicates that 3333 iterations were dropped, which could be due to session not found errors or other issues with session management. To resolve these issues, the recommended architectural and database optimization changes should be implemented.